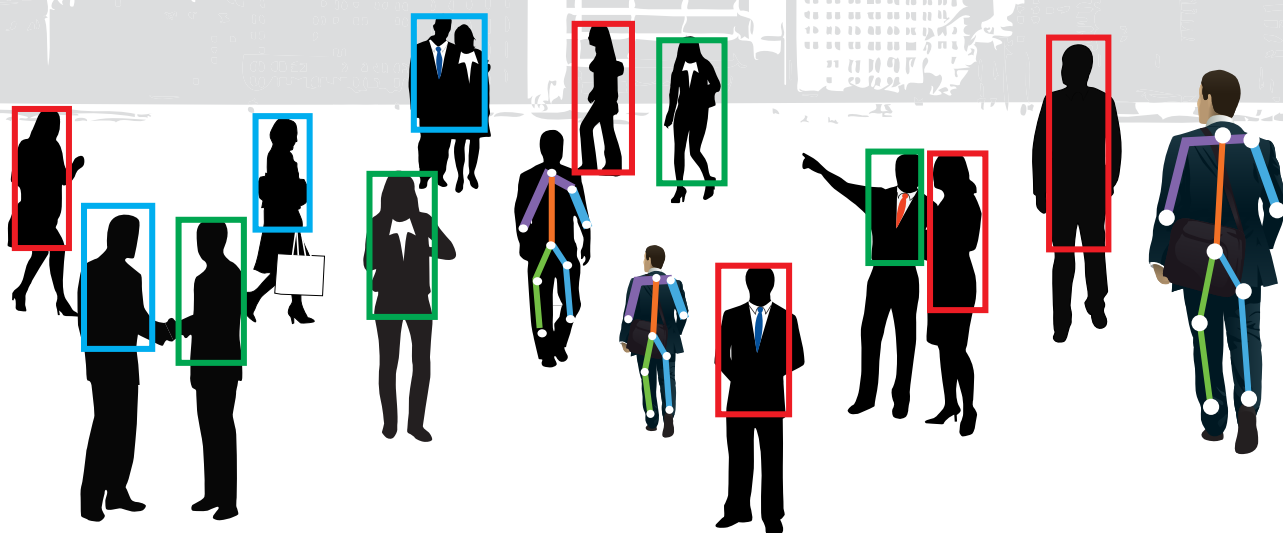


Darknet: The Open Source Framework for Deep Neural Networks

This article gives a quick overview of Darknet and the way it works.



Artificial Neural Networks (ANNs) are a key area of research and application in the field of artificial intelligence. ANNs simulate the algorithms, techniques and strategies adopted by the human brain to process data and information. Robotics, self-driving vehicles, speech recognition, medical image analytics, bioinformatics, natural language processing (NLP), real-time image processing and many other applications make use of such algorithms.

Nowadays, deep learning is widely used for advanced applications of image and video processing with high performance levels. Deep learning neural networks make use of the higher levels of accuracy in prediction and dynamic data analysis, and are now being widely used as an implementation of ANNs.

Free and open source libraries for deep neural networks

Table 1 lists the key libraries and frameworks for implementing deep learning and advanced neural networks.

Library

URL

Darknet	pjreddie.com/darknet/
TensorFlow	tensorflow.org
Keras	keras.io
Caffe	caffe.berkeleyvision.org/
PyTorch	pytorch.org
MXNet	mxnet.apache.org
CNTK	cntk.ai/
Fast.ai	fast.ai/
Deeplearning4j	deeplearning4j.org/
Theano	github.com/Theano/Theano
PaddlePaddle	github.com/PaddlePaddle/Paddle
Sonnet	github.com/deepmind/sonnet
Dlib	dlib.net
Chainer	chainer.org
BigDL	bigdl-project.github.io/
Dynet	dynet.io/

Table 1